

Sensitivity Analyses for Missing Not at Random Data in Body Donor Program Studies

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Problem

- **Context:** Missing Not at Random (MNAR) data, a challenge in observational and experimental studies due to unobserved factors, can bias analyses and compromise decision-making [1].
- **Motivation:** Sensitivity analyses assess robustness to assumptions about missing data, especially MNAR [2]. Most sensitivity analyses focus on missingness in the outcomes, with less emphasis on missingness in categorical covariates.
- **Objective:** How to evaluate the robustness of models with MNAR categorical covariates?

Setting

Body Donor Acceptance Modeling:

- **Model:** Acceptance $\sim$ Education + Occupation + Sex + County Median Income + Age at Registration
- **Interested in:** Predicting acceptance outcomes
- **Importance:** Incorrect predictions may lead to urgent funeral plans or lost full-body donations.
- **Measure:** Prediction accuracy
- **Missingness:** Missingness in education and occupation may influence prediction accuracy.

Results

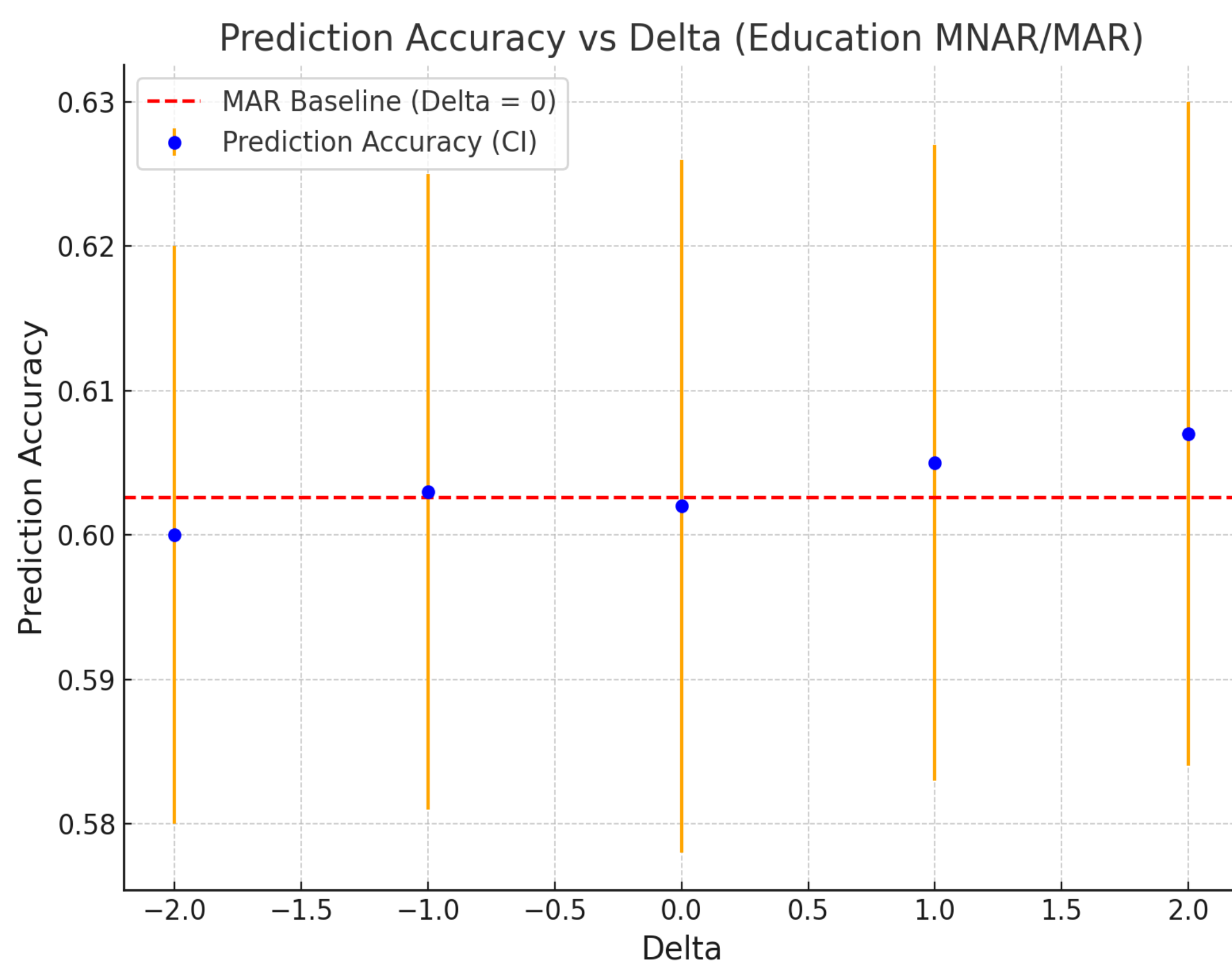


Fig. 1: Education MNAR/MAR

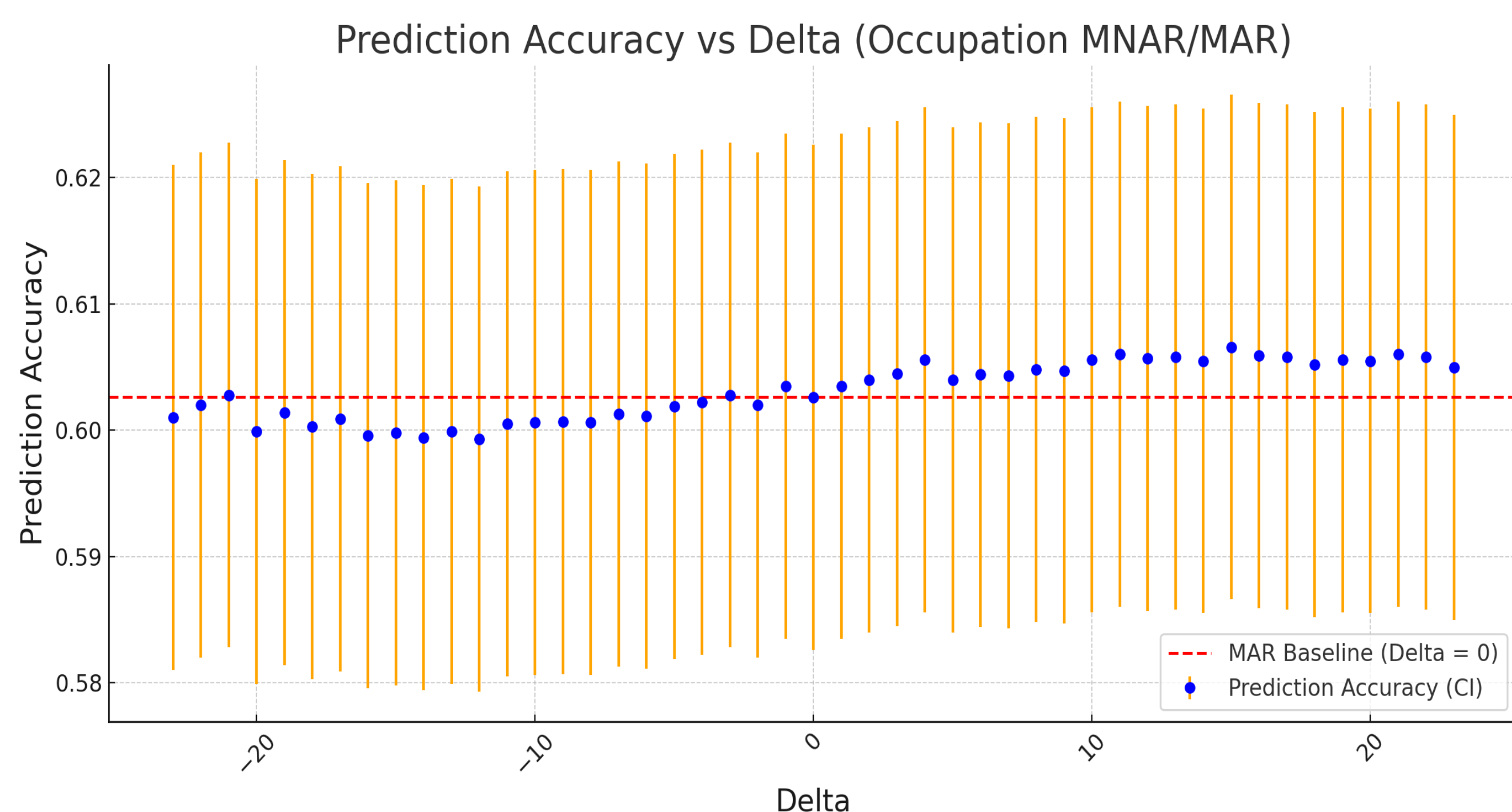


Fig. 2: Occupation MNAR/MAR

Sensitivity Analysis Framework

Logistic Regression Model:

$$P(Y = 1 | X_M = x_M, X_N = x_N) = \beta_0 + \beta_M x_M + \beta_N x_N$$

- X_M : Covariates with missingness.
- X_N : Covariates without missingness.
- Base model fitted with complete data.
- Assumption: no interaction effects.

Ranking:

- Assign ranks $k \in \mathbb{N}$ to each category X_{mi} of X_m based on the order of β_{mi} from the complete data model.
- Use these ranks for subsequent imputations and modeling.
- Assumption: linear relationship between k and Y .

Xedu		Xocc	
A	β_A	a	β_a
B	β_B	b	β_b
C	β_C	c	β_c

Ranking

Xedu		Xocc	
1	1	3	3
2	2	2	2
3	3	1	1

Fig. 3: Coefficient-Based Ranking Framework Example

Missing at Random (MAR):

- Impute all missing covariates under MAR.
- Estimate Y using the regression model for the MAR dataset.

Missing Not at Random (MNAR):

- Evaluate one covariate at a time.
- Define Sensitivity parameter: $\delta = X_{m,mis} - X_{m,imp}$ [2]
- Shift MAR-imputed values by δ to simulate MNAR scenarios [2].
- Estimate Y using the regression model for each MNAR dataset.

edu	occ	edu	occ	edu	occ
NA	3	2	3	2 + 1	3
NA	NA	3	1	3 (max)	1
1	2	1	2	1	2
3	NA	3	1	3	1

Fig. 4: Imputation and Sensitivity Adjustment Workflow Example

Remarks

While modifying multiply-imputed data to reflect departures from MAR is practical and widely used [3], it makes MNAR depend on MAR. This reliance may introduce bias if MAR imputations poorly approximate the true mechanism. This is a flaw in current Multiple Imputation Pattern-Mixture Models. More nuanced approaches should model MNAR mechanisms independently.

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References

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- [2] National Research Council (US) Panel on Handling Missing Data in Clinical Trials. *The Prevention and Treatment of Missing Data in Clinical Trials*. Washington (DC): National Academies Press (US), 2010.
- [3] B. Leurent et al. "Sensitivity Analysis for Not-at-Random Missing Data in Trial-Based Cost-Effectiveness Analysis: A Tutorial". In: *Pharmacoeconomics* 36.8 (Aug. 2018). Erratum in: *Pharmacoeconomics*. 2019 Jul; 37(7): 971. doi: 10.1007/s40273-019-00776-7, pp. 889–901. DOI: 10.1007/s40273-018-0650-5.